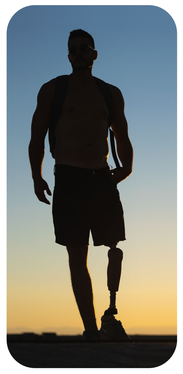
Leveraging Regression-Based Methods for the Design of Parallel Elastic Actuators in Lower-Limb Prosthesis

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Background

- There are currently **2.1 million** people in the US living with **limb loss**
- Globally there are over 1 million amputations each year [1]
- **70%** of prosthetic devices in use are for **below-the-knee** amputations [1]



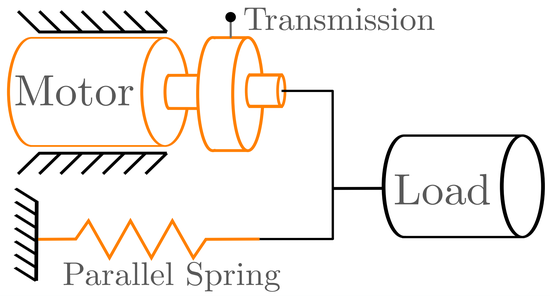


A render of the University of Michigan's OSL [2]

- Our lab specializes in **compliant mechanisms**, which **bend and deform** to store/release energy
- These are **significant advantages** for powered and passive **prosthesis**

Parallel Elastic Actuators

- **Parallel Elastic Actuators** (PEAs): any system with an elastic element (springs) in parallel with a powered actuator (motor)
- PEA's can **reduce power** and load requirements, **extend runtime**, and enable **cheaper actuators**.
- The stiffness is dependent on the **load/deflection profile** for the task
- There is a range of stiffnesses that are beneficial



Regression

Summary

- **Optimal stiffness** for a PEA can obtained by **minimizing** an objective, such as **RMS Torque**
- Usually computed via convex optimization [3]
- We discovered **Weighted Least Squares Regression** (WLSR) is a valid approach
  - Produces a **single optimal** spring
  - Provides **visual intuition**
  - Maintains a **simple** mathematical formulation
  - Enables the design of **higher-complexity** elastic elements, such as **piecewise springs**

Objective

$$min: \sum_{i=1}^{n} w_i (y_i - \tau_i)^2$$

(RMS Torque Minimization as Weighted Regression)

System Inputs

Task Definition (y_i = *walking, stairs*)

Actuator Dynamics $J_m, \mu, \eta, k_m, k_t, \Omega$

Preprocessing

Discretize trajectory ($i = 1 \dots n$)

Weight by duration ($w_i = \Delta t_i$)

Select stiffness model (τ_i = *piecewise, polynomial*)

Output

Unique Optimal Stiffness Coefficients

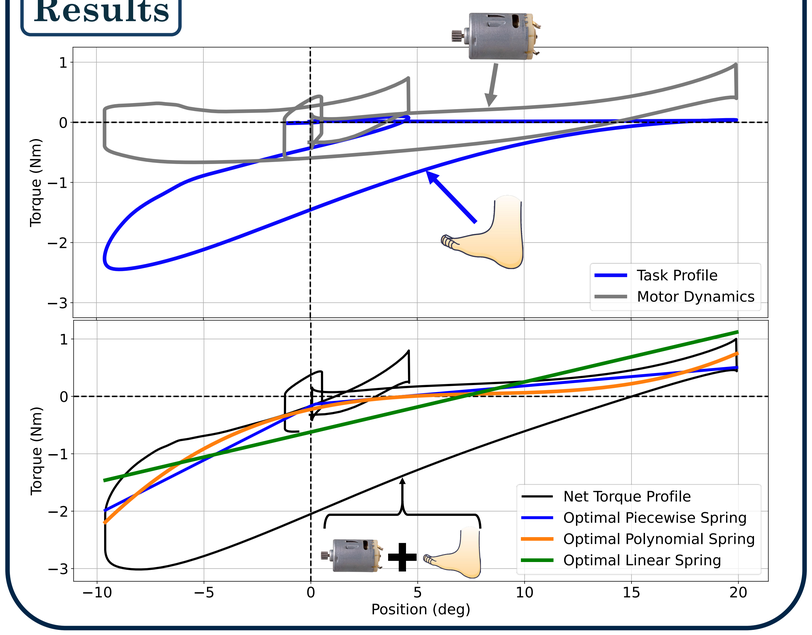
Piecewise Linear Stiffness

$$\tau_s = \begin{cases} A_1 x + B_1; x \le C \\ A_2 x + B_2; x > C \end{cases}$$

Polynomial Stiffness

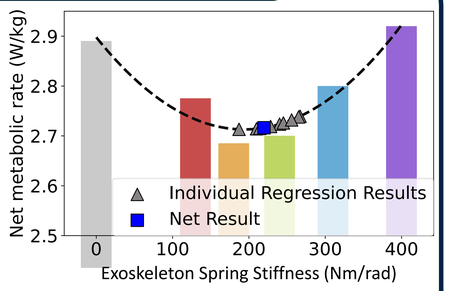
$$\tau_s = A_n x^{n-1} + \cdots A_1$$

Results



Validation

External Dataset



- Ankle walking data from Collins et al. [4] was used to **evaluate** WLSR on a **different dataset**
- WLSR reported an overall stiffness of **219** Nm/rad, within the **180** and **240** Nm/rad reported as **metabolically optimal**
- Subject-specific regressions yield spring constants in the range of **187**-**267** Nm/rad, indicating that **WLSR remains effective** for alternate datasets

Piecewise Spring



- **Future validation** of the regression will replicate a torque/deflection profile **via motors**, connect a regression-optimized **piecewise spring** in parallel, and **measure RMS Torque** reduction
- This will be performed by our collaborators at the University of Notre Dame.

Acknowledgements & Sources

We acknowledge funding from the National Science Foundation under Grant 2344766. This work is in collaboration with the University of Notre Dame.

[1] Pro Medical East. "29 Limb Loss Statistics, Facts & Demographics." *Pro Medical East,* 31 Mar. 2023, promedeast.com/limb-loss-statistics/.
[2] opensourceleg.org. Accessed 30 Oct. 2025.
All uncaptioned images are our own or sourced from Adobe Stock (Educational License).
[3] K. Yang, M. Dickens, J. Schmiedeler, and E. Bolivar-Nieto, "Extending the benefits of parallel elasticity across multiple actuation tasks: A geometric and optimization-based approach," pp. 1–11
[4] S. H. Collins, M. B. Wiggin, and G. S. Sawicki, "Reducing the energy cost of human walking using an unpowered exoskeleton," vol. 522, no. 7555 pp. 212–215.